Mathematical Foundations of Political Methodology

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- Derivative
- Derivative and Continuity
- Derivative Rules
- Mean Value Theorem
- Second Derivatives

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Derivative: Motivation

- Derivatives describe the rates of change in functions
- Foundational across a lot of work in Political Science.
- A special limit
- In this lecture, we will cover three broad ideas
  - Geometric interpretation/intuition
  - Formulas/Algebra derivatives
  - Famous theorems

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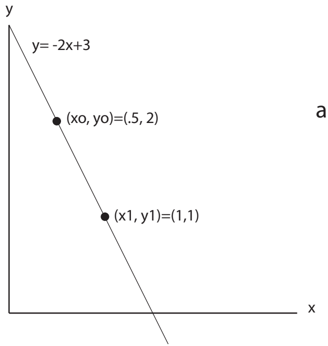
Line and Slope

A line can be represented by the equation $y = a * x + b$, with a being the slope and b the $y-$intercept

For any points on the line, (x_o, y_o) and (x_1, y_1), we have $a = \frac{y_1 - y_o}{x_1 - x_o}$



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=(1-2)/(1-.5)
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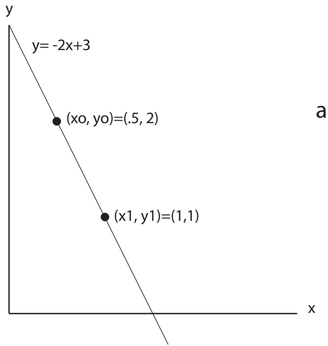

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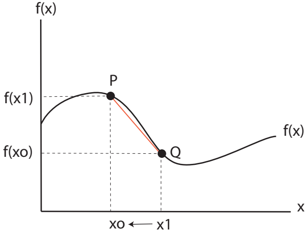
Slope of a Curve

The slope of a curve, $f(x)$, at point P is the limit of the slope of the line between Q and P as $Q \to P$



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The *slope of the curve* $f(x)$ at point P is the limit of the slope of the line between Q and P as $Q \to P$

$P = (x_o, f(x_o))$ and $Q = (x_1, f(x_1))$; we'll write $x_1 = x_o + h$ so that we have $x_1 \to x_o$ as $h \to 0$

We now have

$$P = (x_o, f(x_o)) \text{ and } Q = (x_o + h, f(x_o + h))$$

To find the slope of $f(x)$ at point P we want to find the slope of the line between P and Q as $h \to 0$. This slope is:

$$\frac{y_1 - y_o}{x_1 - x_o} = \frac{f(x_o + h) - f(x_o)}{x_o + h - x_o}$$

As $h \to 0$, this quotient will give us derivative of $f(x)$ at x_o.

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Derivative: Formal Definition

Suppose $f : \Re \rightarrow \Re$. Consider a measure rate of change at a point x_0 with a function $R(x)$,

$$R(x) = \frac{f(x) - f(x_0)}{x - x_0}$$

- $R(x)$ defines the rate of change.
- A derivative will examine what happens with a small perturbation at x_0

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Derivative Definition

Definition

Let $f : \Re \to \Re$. *If the limit*

$$\lim_{x \to x_0} R(x) = \frac{f(x) - f(x_0)}{x - x_0} = f^{'}(x_0)$$

exists then we say that f *is differentiable at* x_0. *If* $f^{'}(x_0)$ *exists for all* $x \in$ *Domain, then we say that* f *is differentiable.*

Derivative Examples

- Suppose $f(x) = x^2$ and consider $x_0 = 1$. Then,

$$\begin{aligned}
\lim_{x \to 1} R(x) &= \lim_{x \to 1} \frac{x^2 - 1^2}{x - 1} \\
&= \lim_{x \to 1} \frac{(x-1)(x+1)}{x-1} \\
&= \lim_{x \to 1} x + 1 \\
&= 2
\end{aligned}$$

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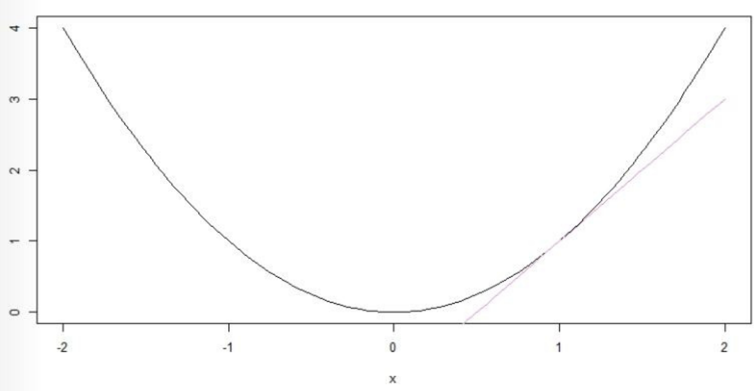

Derivative Examples

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$$\lim_{x \to 0} R(x) = \lim_{x \to 0} \frac{|x|}{x}$$

$\lim_{x \to 0^-} R(x) = -1$, but $\lim_{x \to 0^+} R(x) = 1$.
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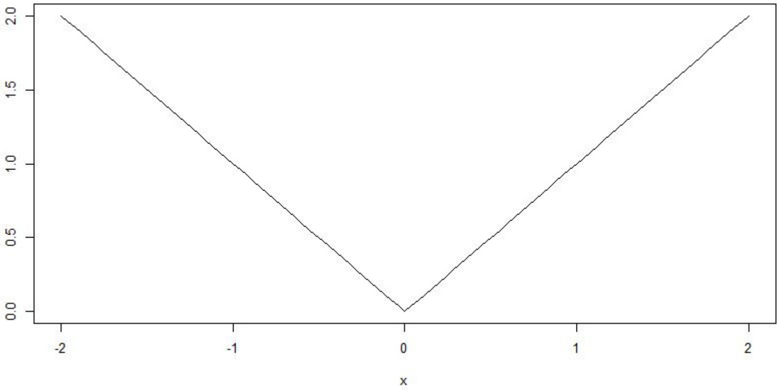

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Derivative and Continuity

Previous example that $|x|$ is continuous but not differentiable at 0 suggests that differentiability is a stronger condition than continuity. Then, a guess would be differentiability imply continuity.

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Proof.

Notice that,

$$\begin{aligned} f(x) &= \frac{f(x) - f(x_0)}{x - x_0}(x - x_0) + f(x_0) \\ &= R(x)(x - x_0) + f(x_0) \end{aligned}$$

If $f(x)$ is continuous at x_0 then, $\lim_{x \rightarrow x_0} f(x) = f(x_0)$.

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Useful Derivative Rules

Suppose a is some constant, $f(x)$ and $g(x)$ are functions

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f(x) = x \quad &; \quad f^{'}(x) = 1 \\
f(x) = ax^{k} \quad &; \quad f^{'}(x) = (a)(k)x^{k-1} \\
f(x) = e^{x} \quad &; \quad f^{'}(x) = e^{x} \\
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Useful Derivative Rules

Suppose a is some constant, $f(x)$ and $g(x)$ are functions

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Derivative in Political Economy

- The policy space is $\mathbb{R}$
- A person (voter?) has an "ideal point" $p \in \mathbb{R}$
- Person prefers policies that are closer to his/her ideal point: receives utility from policy x that is $u(x) = -|x - p|$
- Since this isn't differentiable (and for other reasons), we may assume that

$$u(x) = -(x - p)^2$$
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Algebra of Derivatives

Theorem

Suppose $f : \Re \rightarrow \Re$ *and* $g : \Re \rightarrow \Re$ *and both are differentiable at* $x_0 \in \Re$. *Then,*

i) $h(x) = f(x) + g(x)$ *is differentiable at* x_0 *and*

$$h^{'}(x_0) = f^{'}(x_0) + g^{'}(x_0)$$

ii) $h(x) = f(x)g(x)$ *is differentiable at* x_0 *and*

$$h^{'}(x_0) = f^{'}(x_0)g(x_0) + g^{'}(x_0)f(x_0)$$

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Algebra of Derivatives: Examples

1) $f(x) = x^3 + 5x^2 + 4x$

$$\begin{aligned} f'(x) &= 3x^2 + 5 * 2x + 4 \\ &= 3x^2 + 10x + 4 \end{aligned}$$

2) $f(x) = sin(x)x^3$

$$\begin{aligned} f'(x) &= cos(X)x^3 + sin(x)3x^2 \\ &= cos(X)x^3 + 3sin(x)x^2 \end{aligned}$$

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Chain Rule

Common to have functions in functions

$$
\begin{aligned}
f(x) &= \frac{e^{-\frac{(x-\mu)^2}{2\sigma^2}}}{\sqrt{2\pi}} \\
&= \frac{f(g(x))}{\sqrt{2\pi}}
\end{aligned}
$$

where $f(x) = e^{-g(x)}$, $g(x) = \frac{(x-\mu)^2}{2\sigma^2}$

To deal with this, we use the chain rule

Theorem

Suppose $g : \Re \rightarrow \Re$ and $f : \Re \rightarrow \Re$. Suppose both $f(x)$ and $g(x)$ are differentiable at x_0. Define $h(x) = g(f(x))$. Then,

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- $h(x) = e^{2x}$. $g(x) = e^x$. $f(x) = 2x$. So $h(x) = g(f(x)) = g(2x) = e^{2x}$. Taking derivatives, we have

$$h^{'}(x) = g^{'}(f(x))f^{'}(x) = e^{2x}2$$

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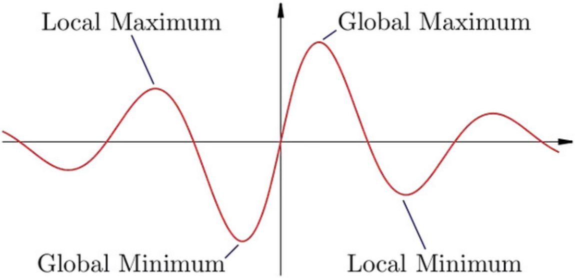
Relative Maxima, Minima and Derivatives



Theorem

Suppose $f : [a, b] \to \Re$. *Suppose* f *has a relative maxima or minima on* (a, b) *and call that* $c \in (a, b)$. *Then* $f^{'}(c) = 0$.

Relative Maxima, Minima and Derivatives

Theorem

Rolle's Theorem Suppose $f : [a, b] \to \Re$ and f is continuous on $[a, b]$ and differentiable on (a, b). Then if $f(a) = f(b) = 0$, there is $c \in (a, b)$ such that $f^{'}(c) = 0$.

Proof Intuition: Consider (WLOG) a relative maximum c. Consider the left-hand and right-hand limits

$$\begin{aligned}
\lim_{x \to c^-} \frac{f(x) - f(c)}{x - c} &\geq 0 \\
\lim_{x \to c^+} \frac{f(x) - f(c)}{x - c} &\leq 0
\end{aligned}$$

Theorem

Rolle's Theorem Suppose $f : [a, b] \rightarrow \Re$ *and* f *is continuous on* $[a, b]$ *and differentiable on* (a, b). *Then if* $f(a) = f(b) = 0$, *there is* $c \in (a, b)$ *such that* $f^{'}(c) = 0$.

But we also know that

$$\begin{aligned} \lim_{x \rightarrow c^{-}} \frac{f(x) - f(c)}{x - c} &= f^{'}(c) \\ \lim_{x \rightarrow c^{+}} \frac{f(x) - f(c)}{x - c} &= f^{'}(c) \end{aligned}$$

The only way, then, that $\lim_{x \rightarrow c^{-}} \frac{f(x)-f(c)}{x-c} = \lim_{x \rightarrow c^{+}} \frac{f(x)-f(c)}{x-c}$ is if $f^{'}(c) = 0$.

Mean Value Theorem

Theorem

If $f : [a, b] \rightarrow \Re$ is continuous on $[a, b]$ and differentiable on (a, b), then there is a $c \in (a, b)$ such that

$$f^{'}(c) = \frac{f(b) - f(a)}{b - a}$$

- Derivative
- Derivative and Continuity
- Derivative Rules
- Mean Value Theorem
- Second Derivatives

Second Derivatives

Now, we can extend our definition of derivatives to higher order derivative. Let's first consider second order derivative.

Definition

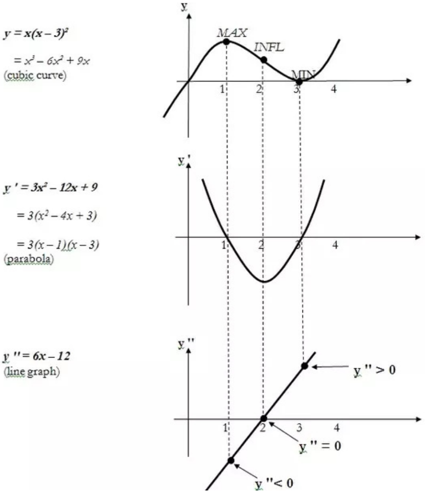
Suppose $f : \Re \rightarrow \Re$ is differentiable. Recall we write this as $f^{'}$ and suppose that $f^{'} : \Re \rightarrow \Re$. Then if the limit,

$$\lim_{x \rightarrow x_0} R(x) = \frac{f^{'}(x) - f^{'}(x_0)}{x - x_0}$$

exists, we call this the second derivative at $x_0, f^{''}(x_0)$.

Second Derivatives

Now, let's see a figure of second derivative.



Example of Second Derivatives

$$\begin{aligned} f(x) &= x \\ f^{'}(x) &= 1 \\ f^{''}(x) &= 0 \end{aligned}$$

Example of Second Derivatives

$$
\begin{aligned}
f(x) &= e^x \\
f^{'}(x) &= e^x \\
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Example of Second Derivatives

$$
\begin{aligned}
f(x) &= \log(x) \\
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f^{''}(x) &= \frac{-1}{x^2}
\end{aligned}
$$

Example of Second Derivatives

$$
\begin{aligned}
f(x) &= \frac{1}{x} \\
f^{'}(x) &= \frac{-1}{x^2} \\
f^{''}(x) &= \frac{2}{x^3}
\end{aligned}
$$

Example of Second Derivatives

$$
\begin{aligned}
f(x) &= -x^2 + 20 \\
f^{'}(x) &= -2x \\
f^{''}(x) &= -2
\end{aligned}
$$